

Alex Sjothun

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Summary

Senior data-focused software engineer with 8+ years designing and delivering analytics platforms, data lakehouse solutions, and dimensional models on Azure and AWS. Deep experience with Azure Data Factory, ADLS, Azure Databricks, Spark, Synapse Analytics, Azure SQL Database, and Fabric, combined with advanced SQL and metadata-driven orchestration patterns. Proven track record leading data engineering projects end to end, from requirements discovery and architecture through implementation, documentation, and mentoring. Strong consultant mindset with a focus on data quality, security, and scalable patterns that drive measurable business outcomes for stakeholders and clients.

Skills

Programming Scripting: Python, SQL, PySpark, Scala, T-SQL, Bash, PowerShell
Data Engineering Warehousing: Dimensional modeling, Star schema design, ETL/ELT pipelines, Metadata-driven orchestration, Data lakehouse architectures, Data migration, Data integration, Real-time and near-real-time data processing
Azure Data Platform: Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage (ADLS Gen2), Azure SQL Database, Microsoft Fabric, Azure Functions, Azure Key Vault
Cloud Infrastructure: Amazon Web Services (AWS), Azure networking and security, Data lakes and data warehouses, Infrastructure as code, Monitoring and alerting, Cost optimization
Data Management Governance: Data quality frameworks, Data reliability and observability, Data security and access control, Row-level and column-level security, Data cataloging and lineage, JSON normalization and flattening, API data ingestion (OAuth 2)
Analytics BI Ecosystem: Power BI, SSAS Tabular concepts, Semantic models, KPI and metric design, Reporting performance tuning
Architecture Leadership: Solution architecture, Technical roadmap definition, Technical mentoring, Cross-functional collaboration, Workshop facilitation, Technical documentation and diagramming, Agile delivery, Stakeholder management
Tools Dev Practices: Git, CI/CD for data pipelines, Testing for data pipelines, Code review, Design patterns, Performance optimization

Experience

Sep 2022 – Feb 2026 (3 years 5 months)

Amazon Web Service (AWS) - Senior Software Engineer, Data

- Led architecture of a multi-tenant analytics platform using Azure Data Factory, ADLS, and Synapse Analytics, enabling governed self-service reporting for enterprise customers.
- Designed and implemented metadata-driven orchestration patterns in Azure Data Factory and Databricks, reducing new pipeline development time by over forty percent across teams.
- Built robust star schemas and dimensional models in Azure SQL Database and Synapse dedicated pools, optimizing for Power BI semantic layers and high-concurrency reporting workloads.
- Implemented a lakehouse pattern combining ADLS, Delta tables in Azure Databricks, and curated warehouse layers, balancing flexibility of data lakes with warehouse reliability and governance.
- Developed complex PySpark jobs in Azure Databricks to standardize, flatten, and normalize nested JSON from diverse APIs into SQL-ready structures for downstream analytics.
- Integrated multiple OAuth 2 secured REST APIs into the platform, building reusable ingestion frameworks with automated schema drift handling and data quality validations.
- Collaborated closely with BI developers, Azure cloud administrators, and product managers to align data models and pipelines with business KPIs and regulatory requirements.
- Introduced data quality and observability checks using custom Python libraries and pipeline-level alerts, significantly improving trust in metrics and reducing incident resolution time.

- Partnered with security architects to implement row-level and column-level security, encryption at rest, and key management via Azure Key Vault across all analytics environments.
- Led design workshops with strategic customers in the cloud services domain, translating business objectives into actionable data roadmaps and phased implementation plans.
- Mentored a distributed team of data engineers, establishing coding standards, review practices, and reusable templates that elevated consistency across Azure data projects.
- Championed comprehensive documentation and architecture diagramming using tools like Azure Architecture Center patterns, improving onboarding speed for new engineers.
- Optimized SQL and Spark transformations for large fact tables, tuning partitioning, indexing, and caching strategies to meet strict SLAs for interactive analytics queries.
- Coordinated end-to-end delivery of complex data migration projects into Azure, managing scope, risks, and stakeholder expectations while consistently meeting launch timelines.
- Collaborated with AWS service teams to ensure hybrid interoperability, enabling customers to integrate Azure-based analytics with existing AWS data sources and workloads.

Oct 2020 – Aug 2022 (1 year and 10months)

Thread - Senior Data Engineer

- Architected a greenfield customer analytics stack on Azure using ADLS, Azure Data Factory, and Azure SQL Database, enabling unified reporting across ecommerce and marketing systems.
- Designed dimensional models and star schemas for customer, product, and order domains, aligning with Kimball principles to support flexible slicing and high-performance dashboards.
- Implemented metadata-driven ETL frameworks in Azure Data Factory and Python, allowing analysts to onboard new sources by configuration rather than custom pipeline code.
- Built PySpark jobs on Azure Databricks to process high-volume clickstream and transaction data, supporting near-real-time insights into user behavior for the retail business.
- Developed robust SQL-based transformation layers with reusable views and stored procedures, encapsulating complex business rules and ensuring consistent KPI definitions.
- Integrated third-party SaaS and advertising APIs secured with OAuth 2, flattening nested JSON payloads into normalized relational structures for downstream modeling and reporting.
- Collaborated with Power BI developers to design semantic models and optimize DAX queries, ensuring responsive dashboards for merchandising and marketing stakeholders.
- Partnered with security and compliance teams to implement data masking, role-based access, and audit logging across the analytics platform, meeting industry requirements.
- Led cross-functional design sessions with product managers and business analysts, capturing requirements and translating them into backlog items and technical specifications.
- Mentored mid-level engineers on Azure data engineering best practices, code review discipline, and effective use of Databricks notebooks for collaborative development.
- Documented end-to-end data flows, lineage diagrams, and runbooks, becoming the go-to resource for understanding platform behavior and troubleshooting complex issues.
- Collaborated with operations leadership in the ecommerce domain to prioritize data initiatives that improved inventory accuracy, campaign attribution, and customer lifetime value.

Oct 2017 – Sep 2020 (3 years)

Newsela – Software Engineer

- Developed backend services in Python and SQL to support analytics features for the education platform, focusing on student engagement and content performance metrics.
- Designed and maintained ETL jobs that consolidated event logs and transactional data into a reporting database, enabling educators to access timely classroom insights.
- Implemented dimensional-style tables and summary aggregates to support dashboards used by school

administrators to track reading progress and curriculum adoption.

- Collaborated with data analysts to refine SQL queries and views, ensuring consistent definitions of key metrics across product analytics and customer reporting surfaces.
- Built integrations to external education APIs, handling OAuth 2 authentication and transforming nested JSON responses into relational tables for downstream analysis.
- Worked with product managers and curriculum specialists to understand reporting needs in the K–12 education domain and translate them into technical data requirements.
- Improved performance of high-traffic reporting endpoints by optimizing SQL, adding appropriate indexes, and caching frequently accessed aggregates in the application tier.
- Participated in code reviews and design discussions, advocating for clear data contracts and reliable schema evolution practices across microservices and data stores.
- Contributed to internal documentation of data flows, schema diagrams, and onboarding guides, helping new engineers quickly understand the analytics-related components.
- Collaborated with DevOps and infrastructure teams to ensure data jobs were monitored, logged, and recoverable, reducing downtime for critical education reporting features.

Education

2013 - 2017

University of Texas at Austin, Austin, TX

Bachelor's degree, Computer Science

Certifications

AWS SimuLearn: Generative AI Architect

AWS SimuLearn: Machine Learning

AWS SimuLearn: Solutions Architect